

Towards Image Design Space Exploration in Spreadsheets with LLM Formulae

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	A	B	C	D	E	F	G
1				14770	66111	17154	7543
2		City in ruins. Post-apocalyptic, crumbling buildings.	=ALTERNATIVES(B2) City in ruins, view from a broken window. Burnt sky.	=IMAGE(TTI(\$C2 , D\$1))			
3			Post-apocalyptic highway, vehicle husks. Burnt sky.				
4		Post-apocalyptic scene objects	City behind walls, sewer draining into ocean.				
5		=GPT_LIST(B4) abandoned car	=B5 & " in a " & C2 abandoned car in a city [...]	=IMAGE(TTI(\$C5 , D\$1))			
6		torn billboard	torn billboard in a city [...]				
7		broken streetlight	broken streetlight in a city [...]				
8		rubble pile	rubble pile in a city [...]				

Figure 1: DREAMSHEETS enables rapid exploration of Text-to-Image prompts in a spreadsheet. An initial prompt (cell B2) is expanded using =ALTERNATIVES into column (C2:C4). With a row of seeds (D1:G1), this yields a set of images (D2:G4). Further exploration is enabled by =GPT_LIST (B5:B8) and concatenation (&) into new prompts (C5:C8). Gray lozenges show user-entered formulas; dashed lines show cell values containing computed or generated content.

ABSTRACT

Users of Text-to-Image (TTI) models like DALL·E and Stable Diffusion typically engage in a lot of iteration, exploring a design space with two main inputs: (1) prompt text spanning image content and style; and (2) stochastic (e.g., random seeds) and other opaque (e.g., classifier-free guidance) variables. Here, we demo an early prototype interface using a spreadsheet metaphor to enable exploration and display of multiple input changes simultaneously, and affording prompt-crafting using spreadsheet formula construction. New LLM-based functions aid rapid exploration of the prompt text input space, by generating new variations on existing prompts and context-relevant lists of prompt keyword options.

CCS CONCEPTS

• Human-centered computing → Interaction paradigms.

KEYWORDS

generative AI, text to image, design space exploration

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1 INTRODUCTION

Artists and other users of new Text-to-Image (TTI) models like DALL·E [12] and Stable Diffusion [15] are faced with a cacophony of tools with which to create images from text. Despite their variety, these tools typically offer a very straightforward interface: a prompt input, perhaps also a negative prompt input, sliders for hyperparameters, and an output area showing a few output images

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and some saving options. The *process* that users engage in when using prompt-input models, however, is **highly iterative and dynamic**, involving substantial editing and re-editing of prompts and evaluating model outputs [5, 17, 18].

In this demo, we present DREAMSHEETS, an early prototype of a creativity support tool for image generation using text-to-image models, based on the spreadsheet metaphor. Users can rapidly generate prompts, variations on prompts, and explore generated image outputs using a familiar grid layout and a “what-if” interaction model with easy history tracking and versatile support for different styles of knowledge organization and management.

Because the possible input space for TTI models is opaque [1] and massive—comprising prompts, seeds, hyperparameters, and even other images as inputs—users are forced to engage in a challenging search process through this space. Commonly used commercial and open-source tools offer relatively little support around prompt construction and visualizing the capabilities of this space, beyond producing single or small batches of images for a specific point in input space (or neighboring points with different seeds, in the case of batches), but these models are so useful that users engage in *ad hoc* design exploration despite these limitations [5].

2 BACKGROUND & MOTIVATION

Our work builds on prior research that shows that considering many alternatives in parallel can effectively aid design space exploration [3, 13], such as through gallery interfaces [10], tracking exploration history [4, 6], offering suggestions for possible input shifts [10], and effective organization [9, 14, 16]. Spreadsheets enable many of these abilities, while commonly-used TTI tools lack these supportive capabilities.

At the surface, prompting can appear as straightforward as instructing a human, but crafting effective prompts can be a challenge [7, 18]. Choosing the right language to achieve desirable visual results in these prompt-based interactions can be difficult, presenting challenges for users seeking to reliably incorporate TTI systems into their exploratory workflows and accomplish creative goals [18]; this has motivated online user communities [11] and researchers to develop and investigate new prompting techniques and tools [2, 7, 8].

3 PROMPT & IMAGE EXPLORATION WITH DREAMSHEETS

DREAMSHEETS leverages the spreadsheet model in support of iteration and exploration of the TTI generation prompt input space, support that is largely missing from commonly-used tools. The features of DREAMSHEETS are embedded within a spreadsheet (built on Google Sheets) that recomputes and re-renders images, allowing drag-based “autofill” and other common spreadsheet functionality. DREAMSHEETS offers access to TTI generation as a spreadsheet function that can take the content of other cells in the sheet, including combinations and transformations of multiple cells, as input. These features support the user in efficiently exploring and gaining a sense of how changes across inputs directly influence their generated outputs.

Function Name	Description
TTI(prompt, [seed], [cfg])	Generate an image (at the returned URL) using the given <i>prompt</i> and, optionally, <i>seed</i> and a classifier-free guidance (<i>cfg</i>) parameters.
GPT(prompt)	LLM function for arbitrary <i>prompt</i> .
GPT_LIST(prompt, length)	Populates <i>length</i> cells in a row (or column) with words/phrases of type <i>prompt</i> .
LIST_COMPLETION(prompt)	Like GPT_LIST, but <i>prompt</i> is a list of items, rather than a description.
SYNONYMS(prompt)	Generates a list of synonyms.
ANTONYMS(prompt)	Generates a list of antonyms.
DIVERGENTS(prompt)	Generates “divergent” words.
ALTERNATIVES(prompt)	Generates a list of alternative wordings for <i>prompt</i> .
EMBELLISH(prompt)	Generates an embellished alternative to <i>prompt</i> , commonly using more specific or detailed words.

Table 1: The LLM-based functions available in our prototype. Each list-producing function additionally has an `_t` alternative (e.g., `SYNONYMS_t`) that transposes the output into a column.

To aid prompt exploration, our prototype also includes a set of LLM-based functions for manipulating prompts directly, including GPT_LIST and LIST_COMPLETION for generating or extending a list of items of a certain description, EMBELLISH to add detail, and ALTERNATIVES to generate multiple variations of a seed prompt (see Table 1 for a full list).

4 USAGE SCENARIO

To motivate our prototype, consider the following scenario:

Paula T. Artist works for a medium-size movie studio. She wants to create some concept art for an upcoming film she plans to pitch about a dystopian vision of the Pacific Northwest region within the United States. Inspired by an initial image of a post-apocalyptic cityscape (Fig. 1, cell B3), Paula wants to generate a few complementary images to use in storyboarding her film. She’s more interested in the aesthetic feel of these images, wanting to convey a certain mood, rather than any specific content—in others words, Paula does not have a specific image in mind that she is trying to produce.

Getting Started. Paula begins by brainstorming ideas for different scenes in the movie. Based on her ideas, she creates a written description that she can use as a prompt through the TTI() function: City in ruins. Post-apocalyptic, crumbling buildings. She uses a cell reference (e.g., B2) to convert her text prompt into an image URL, which is then displayed directly inside a cell using the built-in IMAGE() function. Because our prototype is built on top of a functional spreadsheet (Google Sheets), cells automatically update

in response to changes in referenced cells— changes to a prompt in one cell are rapidly reflected in changes to any images referencing that cell.

Whole-prompt Variations. Dissatisfied with the initial output, Paula turns to DREAMSHEETS' LLM-based ALTERNATIVES function (Fig. 1, cells C2:C4) to generate new prompts: "City in ruins, view from a broken window. Burnt sky.", "Post-apocalyptic highway, vehicle husks. Burnt sky.", and "City behind walls, sewer draining into ocean." Using a row of random seeds, 14770, 66111, 17154, and 7543 (Fig. 1, cells D1:G1), Paula quickly generates a 2D grid of images based on the cross-product of these prompts and seeds (Fig. 1, cells D2:G4). She does so by first entering the formula =IMAGE(TTI(\$C2, D\$1)), and then dragging the "autofill" handle to fill a 2D region of 12 cells to span the three generated prompts and four seed values.

In a first pass at refining these images, Paula iterates on her prompts a few times to see what variations result, adding a new row for each prompt variation in order to preserve her exploration history.

Word-level Variations. Turning to another LLM-based prompt exploration function, GPT_LIST (Fig. 1, cells B5:B8), Paula explores adding a few post-apocalyptic objects to one of her earlier prompts (Fig. 1, cells C5:C8) using string concatenation (&). Calling GPT_LIST with a reference to the string "Post-apocalyptic scene objects" returns the values abandoned car, torn billboard, broken streetlight, and rubble pile; using the formula =B5 & " IN A " & C2, Paula programmatically generates new prompts by combining these objects with the prior prompt, e.g., abandoned car in a City in ruins, view from a broken window. Burnt sky.

Iterating & Sharing. Iterating in this fashion, Paula tries a few hundred prompts over the course of about 10-20 minutes, ultimately finding a set of aesthetically consistent images that she feels effectively capture her desired mood.

As a major side benefit, when Paula is satisfied with her generated images and turns to sharing her work with collaborators, the underlying Google Sheet that Paula used can easily be shared. The embedding of her explorations into a collaborative online spreadsheet means that Paula's collaborators can also explore the full history of how Paula came to the prompts and images she ultimately selected, and what images and prompts she considered along the way.

5 DEMONSTRATION SETUP

Our demonstration setup includes a laptop running the prototype interface, configured to include a few sheets' worth of example usage and a tutorial. Visitors to the demo can also use a short URL to access the interface from their own devices.

6 CONCLUSION

We demonstrate a prototype spreadsheet interface for exploring the input space of Text-to-Image models. Built on top of Google Sheets, our prototype offers a number of new functions for automatically generating whole-prompt variations and component word or phrase variations, and thus enabling the construction of prompts using

spreadsheet formulas. In this demonstration, users can explore the prompt design space on our computer or on their own devices.

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